

Cloud Identity and Access Management Architecture documentation

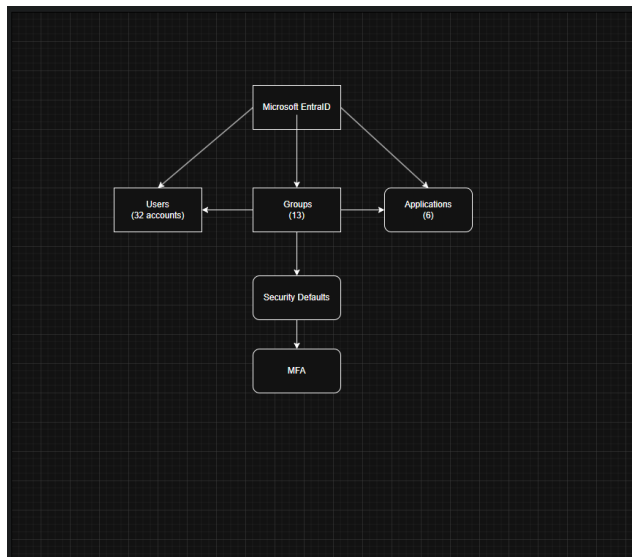
This Identity and Access Management (IAM) architecture project was designed to build hands-on experience with Microsoft Entra ID and core enterprise identity concepts, including user lifecycle management, role-based access control (RBAC), and application access assignment.

The initial objective of the project was to simulate a mid-sized enterprise environment by creating:

- 25 user accounts
- 10 security groups (RBAC roles)
- Multiple enterprise applications with controlled access assignments

Architecture:

This architecture was designed to simulate a mid-sized enterprise Identity and Access Management (IAM) environment using Microsoft Entra ID. The implementation focuses on core IAM principles including user lifecycle management, role-based access control (RBAC), group-based access assignments, enterprise application integration, and multi-factor authentication (MFA). The environment provides a structured identity model that demonstrates how users, groups, and applications can be managed securely and efficiently within a centralized cloud identity platform.



Users:

The 32 user Accounts represent employees across the multiple departments and business. These accounts are to simulate a enterprise level environment to support the testing of role-based access control (RBAC) group memberships, and application access assignments within Microsoft Entra ID.

Toby_Flenderson	Holly_Flax	
Karen_Stafford	Angela_Martin	April Ludgate
Andrew_Dwyer	Ron_Swanson	Todd_Barkley
Tyler_Burke	Ted_Mosby	Jake_Peralta
Jean_Ralphio	Terry_Jeffords	Adrian_Pimento
Ryan_Howard	Pam_Beesley	Doug_Judy
Jerry_Gregich	Garry_Gregich	Charles_Doyle
Chris_Traegar	Ben_Wyatt	Micheal_Scott
Stephen_Gonzales	Oscar_Martinez	Kevin_malone
Gabe_Lewis	Dwight_Shrute	Jim Halpert
Dwight_Shrute	Tom_Haverford	

The screenshot displays the Microsoft Entra ID 'Users' page. It shows a list of 32 user accounts, each with a profile picture, name, email address, and various status indicators. The list is organized into columns: 'Display name', 'User principal name', 'User type', 'Is signed in', 'On-premises sync', 'Identifiable', 'Company name', and 'Creation type'. The users listed include Toby Flenderson, Karen Stafford, Andrew Dwyer, Tyler Burke, Jean Ralphio, Ryan Howard, Jerry Gregich, Chris Traegar, Stephen Gonzales, Gabe Lewis, Dwight Shrute, Holly Flax, Angela Martin, Ron Swanson, Ted Mosby, Terry Jeffords, Pam Beesley, Garry Gregich, Ben Wyatt, Oscar Martinez, Dwight Shrute, Tom Haverford, and others. The interface includes navigation tabs like 'All users', 'New user', and 'Bulk operations'.

32 users found (2 users selected)

☐	Display name ↑	User principal name ↑	User type	Is Agent	On-premises sy...	Identities	Company name	Creation type
☐	 Michael Scott	Michael_Scott@e...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technologies
☐	 Oscar Martin	Oscar_Martin@en...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technologies
☐	 Pamela Beesley	Pam_Beesley@ent...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technologies
☐	 Ron Swanson	Ron_Swanson@en...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technologies
☐	 Ryan Howard	Ryan_Howard@en...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technologies
☐	 Stephen Gonzales	Stephen_Gonzales...		Member	No	No	entraiamproject.onmicrosoft.co	Bug eyed Technologies
☐	 Ted Mosby	Ted_Mosby@entra...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technologies
☒	 Terry Jeffords	Terry_Jeffords@en...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technology
☐	 Toby Flenderson	Toby_Flenderson...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technologies
☐	 Todd Barkley	Todd_Barkley@en...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technologies
☐	 Tom Haverford	Tom_Haverford@...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technologies
☐	 Tyler Burke	Tyler.Burke@entra...		Member	No	No	entraiamproject.onmicrosoft.co	Bug Eyed Technology
☐	 Tyler Burke	890002777_ciat.e...		Member	No	No	ExternalAzureAD	
☐	 Tyler Burke admin	Tyler_Burke_Admi...		Member	No	No	entraiamproject.onmicrosoft.co	BugEyed Technologies

Groups:

The 13 groups represent the departments used in the enterprise environment.

Groups
Policy
HR_managment
Hr
IT_Managment
IAM Admin
IT_Systems/Support
IT_Help_Desk
Networking
Finance_Managment
Finance
Sales_management
Sales
Executives
Red-team contractors
Threat Actors

Search resources, services, and docs (G+)					
Copilot					
890002777@cia.edu					
BURKE-IAM-LAB (ENTRA/AMPRO)					
ups					
...					
New group Download groups Refresh Manage view Delete Got feedback?					
Microsoft Entra has a simpler, integrated experience for managing all your identity and Access Management needs. Try the new Microsoft Entra admin center.					
Search					
Add filter					
Search mode Contains					
13 groups found					
☐	Name	Object id	Group type	Membership type	Email
☐	GRP_Executives	54639c07-730c-445c-933e-f563cd5e0218	Security	Assigned	
☐	GRP_Finance	b55752be-9d2b-43ab-aad2-cc9d2bd3d1af	Security	Assigned	
☐	GRP_Finance_MGT	134c7eb1-f95c-4678-80dd-e0632dd23f52	Security	Assigned	
☐	GRP_HR	72b65e37-e190-4417-8714-7d968b01a3f1	Security	Assigned	
☐	GRP_HR_MGT	aff29c08-3812-4f69-ab6e-24922fa8170b	Security	Assigned	
☐	GRP_IT_Help_Desk	c7844ad1-e4d0-47fc-9434-9a78a05e98cb	Security	Assigned	
☐	GRP_IT_MGT	84725507-f29-460e-b0ea-6a4fcc605a0c	Security	Assigned	
☐	GRP_IT_Systems_Support	4f5e1f6b-38ea-462d-bc9b-fdfb531f9226	Security	Assigned	
☐	GRP_Networking	31d9f44d-23d7-432d-b1d8-3cee44b828cf	Security	Assigned	
☐	GRP_RedTeam_Contracts	ecc7e22b-6a27-44ac-9279-163fe07f3683	Security	Assigned	
☐	GRP_Sales	2065c7db-f38f-4310-8ae7-6fb0f98bfb84	Security	Assigned	
☐	GRP_Sales_MGT	98512135-6c21-4dcd-9f39-ca9b68e52ac4	Security	Assigned	
☐	SEC_IAM_Admins	da29fce0-c5e1-4816-8ad2-a0578e7783aa	Security	Assigned	

Users & Groups: In this section I mapped which user would belong to which group, the light blue colors represent the mixture of multiple roles.

Role-Based Access Control (RBAC) Matrix				
Groups	users			
HR_managment	Toby_Flenderson	Holly_Flax		
Hr	Karen_Stafford	Angela_Martin	April Ludgate	
IT_Managment	Andrew_Dwyer	Ron_Swanson		
IAM Admin	Tyler_Burke	Ted_Mosby		
IT_Systems_and_Sup	Jean_Ralphio	Terry_Jeffords		
IT_Help_Desk	Ryan_Howard	Pam_Beesley		
Networking	Jerry_Gregich	Garry_Gregich	Charles_Doyle	Todd_Barkley
Finance_Managment	Chris_Traegar	Ben_Wyatt		
Finance	Stephen_Gonzales	Oscar_Martinez	Kevin_malone	
Sales_management	Gabe_Lewis	Dwight_Shrute		
Sales	Dwight_Shrute	Tom_Haverford		
Executives	Micheal_Scott	Jim_Halpert		
Contractors/ Supporting Roles				
Red-team contractors	Adrian_Pimento	Jake_Peralta	Tyler_Burke	
Threat Actors	Doug_Judy			

Microsoft Azure

Search resources, services, and docs (G+)

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BURKE-IBM-LAB (ENTRAUNIFIED)

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Preview info

Columns

Preview features

Got feedback?

View, filter, and search applications in your organization that are set up to use your Microsoft Entra tenant as their identity Provider.

The list of applications that are maintained by your organization are in [application registrations](#).

Search by application name, application template ID or object ID

Application type == Enterprise Applications

Application ID starts with

Add filters

6 applications found

Name	Object ID	Application ID	Homepage URL	Created on	Activation status	Certificate Expiry Status	Active Certificate Expir...	Identifier URI (Entity ID)
Salesforce	899ee9b-44c7-4d46-aaca...	a17d8b0-50cd-4fcc-ae92...	https://account.activedirect...	6/8/2026	Activated	-	-	a17d8b0-50cd-4fcc-ae92...
AWS IAM Identity Ce...	afcb682d-6e7d-4290-9cbe...	eb5a06f0-d68c-4e76-bafc...	https://account.activedirect...	6/8/2026	Activated	-	-	eb5a06f0-d68c-4e76-bafc...
Splunk	b5bac9c2-8955-453b-8f30...	7b6f4996-7450-4a30-9f74...	https://account.activedirect...	6/8/2026	Activated	-	-	7b6f4996-7450-4a30-9f74...
ServiceNow	d66e4e0d-e1e1-4abe-8a42...	0230e4a2-b456-41cd-9f58...	https://account.activedirect...	6/8/2026	Activated	-	-	0230e4a2-b456-41cd-9f58...
Netsuite	f127f84-f73f-4c16-8a02-3...	000566af-f79c-4286-a99a...	https://account.activedirect...	6/8/2026	Activated	-	-	000566af-f79c-4286-a99a...
Workday	faafbe3a-5557-43c9-a204-9...	2c02437d-7dca-48c4-aabf...	https://account.activedirect...	6/8/2026	Activated	-	-	2c02437d-7dca-48c4-aabf...

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Self-service

Custom security attributes

Security

Activity

Troubleshooting + Support

+

Add user/group

Edit assignment

Remove assignment

Update credential

Refresh

Manage view

Got feedback?

The application will appear for assigned users within My Apps. Set 'visible to users?' to no in properties to prevent this.

Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#)

First 200 shown, search all users & groups

Display name	Object type	Role assigned
☐ Adrian Pimento	User	User
☐ Andrew Dwyer	User	User
☐ Angela Martin	User	User
☐ April Ludgate	User	User
☐ Ben Wyatt	User	User
☐ Charles Doyle	User	User
☐ Chris Traegar	User	User
☐ Douglas Judy	User	User
☐ Dwight Shrute	User	User
☐ Gabe Lewis	User	User
☐ Garry Gergich	User	User
☐ Holly Flax	User	User

Microsoft Azure

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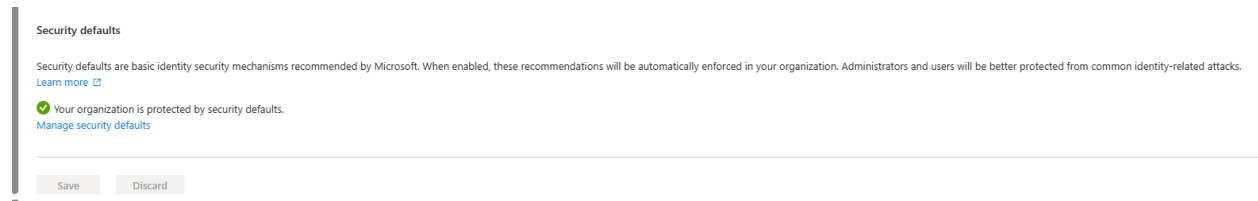
Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#)

First 200 shown, search all users & groups

Display name	Object type	Role assigned
☐ Ben Wyatt	User	User
☐ Chris Traegar	User	User
☐ Jim Halpert	User	User
☐ Kevin Malone	User	User
☐ Michael Scott	User	User
☐ Oscar Martin	User	User
☐ Stephen Gonzales	User	User
☐ Ted Mosby	User	User
☐ Tyler Burke	User	User
☐ Tyler Burke admin	User	User

MFA:

Due to limitations of my EntraID account I cannot perform separate user accounts however I have the Security default active to force MFA on all privilege users



Lessons Learned:

This project provided practical experience with the design and implementation of an Identity and Access Management (IAM) environment using Microsoft Entra ID. Throughout the implementation process, several key IAM concepts were reinforced, including user lifecycle management, role-based access control (RBAC), group-based permission assignment, and enterprise application integration.

One of the most significant lessons learned was the importance of designing access around groups and roles rather than assigning permissions directly to users. This approach improves scalability, simplifies administration, and aligns with industry best practices for identity governance. The project also demonstrated how nested security groups can be used to create more flexible and maintainable access structures.

Working within the limitations of the available Entra ID licensing provided valuable insight into the differences between enterprise IAM design and platform-specific capabilities. While certain advanced features were unavailable, alternative solutions such as Security Defaults and group-based access assignments were successfully implemented to achieve the project's objectives.

The project also highlighted the importance of proper planning and documentation when managing identities, permissions, and application access. As the environment expanded beyond the original scope, maintaining clear relationships between users, groups, and applications became increasingly important.

Final Implementation Summary

During implementation, the environment evolved beyond the original project scope and resulted in:

- 32 user accounts
- 13 security groups
- 6 integrated enterprise applications

The environment included simulated SaaS and cloud-based platforms representing business functions such as Human Resources (HR), Customer Relationship Management (CRM), IT Service Management (ITSM), and cloud infrastructure administration.

This project successfully demonstrated core IAM principles including identity lifecycle management, RBAC design, group-based access control, application assignment, and multi-level group structures within Microsoft Entra ID. Despite licensing limitations, a structured and scalable identity model was designed, implemented, and documented with clear relationships between users, security groups, and application access permissions.

The completed environment serves as a foundation for Phase 2, which will extend identity management into AWS through federation with AWS IAM Identity Center, enabling a centralized multi-cloud identity and access management architecture.

Next Steps: AWS IAM Integration

The next phase of this project will focus on extending the identity architecture into Amazon Web Services (AWS) by integrating Microsoft Entra ID with AWS IAM Identity Center through SAML-based federation. This will enable centralized identity management, allowing users authenticated through Entra ID to securely access AWS resources without requiring separate AWS credentials. The implementation will include role-based access control (RBAC) using AWS IAM roles, user and group mapping between Entra ID and AWS, and the application of least-privilege access principles. This phase will demonstrate a multi-cloud identity management solution and provide hands-on experience with enterprise-grade identity federation and access governance across both Microsoft and AWS cloud platforms.